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3D augmented fundus images for identifying glaucoma via transferred convolutional neural networks

Peipei Wang · Mingyuan Yuan · Yan He · Jiuai Sun

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Abstract
Purpose Glaucoma is a chronic and irreversible retinopathy threatening the vision of millions of patients around the world. Its early diagnosis and treatment can help to prolong the period of sight deterioration from no visual impairment to blindness, whereas the screening and diagnosis of glaucoma in clinical remains challenging because some key assessment criteria like cup-to-disc ratio is limited by subjective analysis and intra- and inter-observer variability. This paper exploits the potential of new augmented image data of the optic nerve head (ONH) combining with the latest deep learning networks to achieve better diagnosis of glaucoma.
Methods This paper explores the potential value of additional three-dimensional topographic map of the optic nerve head proceeded by the latest deep learning approaches, i.e. convolutional neural networks to improve the diagnosis efficiency. Specifically, 3D topography map of the ONH and RGB fundus image has been used to train the transferred AlexNet and VGG-16 networks. The diagnostic performance is compared to those achieved by using the 2D fundus images only.
Results The 3D topographic map of ONH reconstructed from the shape from shading method provides better visualization of the structure of optic cup and disc. These new enhanced dataset was employed to train the proposed deep learning networks and finally achieve diagnostic accuracy of 94.3% which is superior to the networks trained via 2D conventional images.
Conclusion Employing the deep learning neural networks with augmented 3D images can increase the accuracy of automatic separating glaucoma and non-glaucoma fundus images. It may be used as an objective tool in developing computer assisted diagnosis systems for assessment of glaucoma.

Keywords Glaucoma · 3D · Convolutional neural network · Transfer learning

P. Wang · M. Yuan
Department of Radiology, Shanghai University of Medicine and Health Sciences Affliated Zhoupu Hospital, Shanghai 201318, China

P. Wang · J. Sun (✉)
College of Medical Imaging, Shanghai University of Medicine & Health Sciences, Shanghai 201318, China
e-mail: sunja@sumhs.edu.cn

Y. He (✉)
Department of Ophthalmology, The Second Xiangya Hospital, Central South University, Changsha 410011, Hunan Province, China

Y. He
Hunan Clinical Research Center of Ophthalmic Disease, Changsha 410011, Hunan, China

Y. He
Clinical Immunology Center, Central South University, Changsha 410011, Hunan Province, China

Introduction

Glaucoma, a chronic fundus problem caused by degeneration of the retinal optic nerve fibres [1], has become the second leading cause of blindness, with approximately 60 million cases reported worldwide in 2010 and up to 80 million expected in 2020 [2]. What is even worse, over half of these cases have advanced visual field defects when they are presented at first clinical evaluation [3]. Meanwhile, there is no any single measure well-defined for establishing glaucoma progression in clinical, but only the following three approaches are frequently adapted to assist ophthalmologists to make a diagnosis. The first one is the tonometer measuring the intraocular of pressure (IOP) caused by aqueous humour. Although an elevated IOP may cause the optic nerve and visual field being deteriorated, it does not necessarily mean to develop and/or progress into glaucomatous damage as the measurement is often subject to bias or error from various sources. In this case, it is also called ocular hypertension, which differs from glaucoma. Therefore, additional clues such as fundus images shall be brought in to further assess the degree of optic nerve injury [4]. The second approach to detecting glaucoma often identifies and calculates the ratio of cup-to-disc of the ONH and a larger value of the ratio implies glaucoma compared to those smaller ones [5]. Other than identifying the cup-to-disc ratio from fundus images, the third frequently used approach to differentiating glaucomatous damage is to use ophthalmoscopy or retinal tomography to check the thickness of the neuroretinal rim and verify through the principle of ISNT. Generally, the normal neuroretinal rim is the thickest inferiorly and thinnest temporally and generally follows the relationship of inferior (I) $\geq$ Superior (S) $\geq$ Nasal (N) $\geq$ Temporal (T), while glaucomatous damage will cause atrophy along the inferior and superior rims and break down this relationship [6, 7]. The effectiveness of the third approach and the ISNT rule for differentiating glaucomatous damage has being investigated in previous work [8–10]. Comparing these three approaches, diagnosis through fundus images is most convenient and promising because fundus cameras are most accessible and able to provide topological retinal image with good resolution. However, it is sometimes labour-intensive or experience-based when ophthalmologists are required to manually segment the fundus region and accurately identify the border of optic cup and disc. An automatic and objective way of assisting ophthalmologists to diagnose glaucoma are still highly demanded in clinical.

In recent years, computer assisted diagnosis approaches have been proposed to improve the detection of the fundus conditions from fundus photographs, especially when the convolutional neural networks (CNNs) have been emerged [11]. For example, a CNN based on automatic feature learning has been used to classify nuclear cataract [12]; another CNN has been developed to assist the diagnosis of diabetic retinopathy through colour fundus images [13]; a multi-categorical deep neural network has been used to classify 10 types of retinal diseases [14]. However, there appears less work addressing the classification of glaucoma [15–20]. Chen et.al at first proposed a CNN employing four convolutional layers and two fully connect layers for detecting glaucoma with reasonable accuracy [15]. Orlando et al. used a deep neural network pre-trained by a large number of non-glaucoma images to improve the performance of classification [16]. They had noticed the image preprocessing had a significant effect on final classification results. However, they have not further investigated the effects of data augmentation, and the validity of their approach is also expected to be verified on more datasets. Researchers have made a lot of efforts to improve the performance of using CNN to classify glaucoma. Eddine et.al treated the RGB channels separately to form a multimodal machine learning approach to do the glaucoma classification task [17]. Claro et al. tried to use feature extraction algorithm to improve the CNN classification ability of identifying glaucoma, but they only achieved accuracy rate of 93.61% [19]. Marcos et al. proposed a method removing the texture of blood vessels from retinal images, and achieved reasonable results to automatically classify glaucoma and normal retinal conditions [20].

Different from the above work where only colour information of fundus images is used directly, we further exploit the potential of new augmented data, the reconstructed topographic data of the ONH, to achieve better intelligence classification of glaucoma. We implemented and verified various data preprocessing strategies and took advantage of the pre-trained AlexNet and VGG-16 architectures to form transferred new networks by altering their last three fully connect layers. The ONH images used for experiments are obtained from those well-known online databases for training and testing the proposed data augmentation approach and the proposed networks. The performance of classification system is mainly quantified through the criteria including accuracy (ACC), area under curve (AUC) and Cohen's kappa [21].

Methods

Material

Traditionally clinical studies detecting abnormal retinal conditions from fundus image data need to control demographic or imaging variables for validating research hypotheses, however, this work intends to train an end-to-end network system able to generate meaningful diagnostic results from raw photographs with a wide range of diversity. In order to achieve this purpose, the fundus images used in this paper are collected from four publicly available online data sets listed in Table 1. DRIONS-DB database containing 110 retinal fundus images of patients with mean age of 53.0 years old is validated and maintained by the Ophthalmology Service at Miguel Servet hospital, Spain [22]. High-Resolution Fundus (HRF) originally developed for evaluating automatic segmentation algorithms is a public database containing 15 images of healthy patients, 15 images of patients with diabetic retinopathy and 15 images of glaucomatous patients [23]. RIM-ONE dataset is an open retinal fundus images database containing 445 colour images with 251 normal and 194 glaucoma fundus images, respectively [24]. Drishti-GS1 dataset developed for the task of ONH segmentation for automated glaucoma assessment contains 31 normal fundus images and 70 glaucoma fundus images and is managed by the Medical Image Processing (MIP) group, IIIT Hyderabad [25]. The collected images from different sources are presented with a diverse of resolution and compacted format, but all labelled or annotated by more than one medical expert with solid education and experience in ophthalmology.

Table 1 Retinal fundus images used in the paper

Database name	Normal	Glaucoma	Total
DRIONS-DB	50	60	110
HRF-dataset	15	15	30
RIM-ONE	251	194	445
Drishti-GS1	31	70	101
Total	347	339	686

In order to make use of the pre-trained AlexNet and VGG-16 CNN networks, all the ONH images were resized into the format of 227 × 227 and 224 × 224 pixels, respectively. Meanwhile several conventional image augmentation approaches including random scale, random crop, horizontal/vertical flip as well as rotations are used to expand the database for learning and testing the CNN networks, which is called AU ONH data set in the following paper. In addition, a shape from shading (SFS) approach has been used to reconstruct the topography of the ONH, and this part of reconstructed data is called as 3D map data set [27]. As such the fundus data have been doubled and finally we obtain total 686 images including 347 normal and 339 glaucoma images which are then randomly split into 70%, 20% and 10% for the purpose of training, validation and tests, respectively.

All the computation is carried out on a DELL Precision 7810 Tower work station containing an Intel i7 3570 k 3.4 GHz processor, 64 GB of RAM and a Xeon E5-2620 v4 processing unit.

3D reconstruction of ONH

The SFS method used for reconstructing the 3D topography of the ONH was early developed by Horn in the 1970s [26] for recovering the shape of an object from its shading information. The reflection from the interested object is assumed able to be modelled as Lambertian reflection format, i.e. the luminance reflected in all directions is equally distributed. At the same time the surface of the object is assumed smooth and continuous, i.e. the shape of surface has a first-order continuous derivative in the process of restoring the 3D shape of the object. On the premise of satisfying the above conditions, the image brightness of the surface is determined by three parameters, namely, the direction of the light source, the normal vector and the reflectivity of the object surface.

$$E(x, y) = \rho n n_o \quad (1)$$

Where $E(x, y)$ is the pixel intensity of object in the image at point (x, y); ρ is the optical reflectivity of the surface, $n(n_1,n_2,n_3)$ is the unit vector of surface normal at the point, and $\mathbf{n_0}(n_{01},n_{02},n_{03})$ is the direction of the light source. As the ONH is only small part of a whole fundus image, we can treat the illumination as a parallel source. So the brightness function can be simplified as the following expression providing the reflectivity of surface is uniform.

$$E(x,y) = \frac{n_0 \cdot n}{\|n_0\| \cdot \|n\|} = \frac{n_{01}n_1 + n_{02}n_2 + n_{03}n_3}{\sqrt{n_{01}^2 + n_{02}^2 + n_{03}^2}\sqrt{n_1^2 + n_2^2 + n_3^2}} \quad (2)$$

Assuming $z = z(x, y)$ is the surface height of the ONH at the point (x, y), the relation between $z\ (x, y)$ and the gradient is $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$. And the surface normal and illuminance direction can be expressed as $n = (p, q, -1)/\sqrt{p^2 + q^2 + 1}$, $\mathbf{n_0} = (p_0, q_0, -1)/\sqrt{p_0^2 + q_0^2 + 1}$. Substitute n and $\mathbf{n_0}$ into Eq. (2) and the discrete approximations for surface normal p and q are able to be calculated by using the height map $z\ (i, j)$ as:

$$\begin{aligned} p &= \frac{\partial z}{\partial x} = z_{i,j} - z_{i,j-1} \\ q &= \frac{\partial z}{\partial y} = z_{i,j} - z_{i-1,j} \end{aligned} \quad (3)$$

where $i = 0, 1, 2\ldots M - 1$; $j = 0, 1, 2\ldots N - 1$. M and N are the rows and columns of the fundus image. And then the image irradiance equation can be modified into the following difference format:

$$f(z_{i,j}) = E_{i,j} - R(z_{i,j} - z_{i,j-1}, z_{i,j} - z_{i-1,j}) = 0 \quad (4)$$

Next, Taylor expansion is used to modify the Eq. (4).

$$0 = f(z_{i,j}) \approx f(z_{i,j}^{n-1}) + (z_{i,j} - z_{i,j}^{n-1})\frac{\partial f}{\partial z_{i,j}}(z_{i,j}^{n-1}) \quad (5)$$

If the height at the nth iteration is assumed as:

$$z_{i,j}^n = z_{i,j} \quad (6)$$

And the ONH image irradiance equation becomes:

$$z_{i,j}^n = z_{i,j}^{n-1} + \frac{-f(z_{i,j}^{n-1})}{\frac{\partial f}{\partial z_{i,j}}(z_{i,j}^{n-1})} \quad (7)$$

where $\frac{\partial f}{\partial z_{i,j}}(z_{i,j}^{n-1}) = -\left[\frac{p_0+q_0}{\sqrt{p_0^2+q_0^2+1}\sqrt{p^2+q^2+1}} - \frac{(p+q)(p_0p+q_0q+1)}{\sqrt{p_0^2+q_0^2+1}\sqrt{p^2+q^2+1}}\right]$.

Finally, assuming the initial height $z_{i,j}^0 = 0$, the height of the ONH can be calculated iteratively from (7).

Developing glaucoma-net

Compared to traditional pattern recognition approaches, the CNNs emerged in recent years have exhibited a more powerful performance on feature extraction and object classification. In order to have a good performance, they normally require a large number of data to train the network. Although we already have hundreds of fundus images, they are probably not large enough to form a stable network. In order to alleviate the problem of the lack of clinical data, we will take advantage of those well-developed CNNs via a technique called transfer learning, i.e. the task related available images are used to train the last fully connected layers only, while the convolutional layers are used directly from the existing CNNs pretrained for a large scale image classification task. Transfer learning using existing resource allows rapid progress or improved performance in modelling the second task, which has been found successful use in the computer assisted classification applications [28, 29].

The AlexNet and VGG-16 are two commonly accepted CNN architectures with a significant performance in the tasks of classification. Both are trained on tremendous 1.2 million images belonging to over 1000 categories in the well-known ImageNet database. AlexNet contains five convolutional layers, three max pooling layers and two fully connect layers, while the VGG-16 includes thirteen convolutional layers, five max pooling layers and three fully connect layers. In both networks, convolutional layers are used to extract the feature of input image, max pooling layer are used to reduce the dimension of the data. Here we made use of their convolutional layers directly, and only reconfigured the last three layers according to the preprocessed ONH images. The new established networks are called Glaucoma-Net. As a proper parameter setting can significantly affect the final performance of neural network, we finally set the hyperparameters of Weight Learn Rate Factor, Bias Learn Rate Factor and Initial Learn Rate into 5, 5, 0.01, respectively, for this binary classification problem.

Performance evaluation

It is significant to evaluate the network after the deep learning system has been established. In this paper, the evaluation criteria including accuracy (ACC), area under curve (AUC) and Cohen's kappa were employed to test the performance of the Glaucoma-Net system. ROC curve is a graphical plot of sensitivity (SEC) versus its specificity (SPC) and considered as an effective method of evaluating the quality or performance of diagnostic performance too. The sensitivity and specificity are defined as the followings:

$$SEC = \frac{TP}{P} \tag{8}$$

$$SPC = \frac{TN}{N} \tag{9}$$

$$ACC = \frac{TP + TN}{P + N} \tag{10}$$

where TP and TN are the true positive and true negatives prediction of glaucoma; P and N represent positive and negative instances of glaucoma cases. The accuracy is the proportion of the total number of predictions that were correct. The area under the ROC curve (AUC) is the average value of sensitivity for all possible values of specificity and becomes the most popular measure of the overall performance of diagnosis. In the binary classifier, the range of the Kappa value indicates the degree of consistency between the classification results and its true situation. Landis and Koch suggested that the kappa value was considered as acceptable in the range of 0.21–0.40, moderate in the range of 0.41–0.61, greater in the range of 0.41–0.61 and almost perfect in the range of 0.81–1.0 [30]

Results

A normal ONH is shaped like a ring with a pink neuroretinal rim around outside and a central pale shallow indentation in the center. As the region of interest for glaucoma diagnosis, the ONH structure is firstly segmented from colour fundus images. As the ONH with an average diameter around 1.5 mm occupies a relatively small area within the whole back of eye, we increase the resolution of the segmented fundus image in Fig. 1a through the bilinear interpolation method to a standard uniform format to satisfy the requirement of the input images of the networks. The 3D topography map of ONH reconstructed by the SFS method is shown in Fig. 1b. Compared to the original colour image, the optic cup (red curve) and disc (blue curve) can be found more easily from the reconstructed 3D topographic map. It will be verified whether the employment of the 3D topographic data could help to improve the performance of the following classification system.

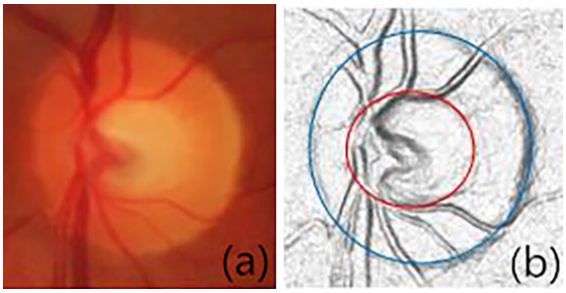

Fig.1 Preprocessed ONH fundus images. (**a**) ONH fundus image; (**b**) 3D topography representation of ONH fundus

The criteria including ROC curves, AUC, SEC, SPC, ACC and Cohen's kappa have been used to quantify the performance of the Glaucoma-Net neural networks transferred from AlexNet and VGG-16. Table 2 and Fig. 2 show the classification performance of Glaucoma-Net tested and validated by using the original colour ONH images, 3D reconstruction of ONH images and augmented colour fundus images, respectively.

Discussion

The results in Table 2 reveal that the Glaucoma-Net transferred from VGG-16 outperforms that transferred from AlexNet in classifying the fundus images of glaucoma and non-glaucoma. Furthermore both 3D reconstruction of ONH image and conventional image augmentation can increase the classification accuracy of the neural networks. Table 2 also shows that either transferred AlexNet or transferred VGG-16 network training by augmentation images achieved good classification performance, and the transferred VGG-16 network demonstrated better performance than the

Table 2 Evaluate Glaucoma-Net system through AUC, SPC and SEC

Dataset	AUC	SPC	SEC	ACC	Cohen's kappa
ONH-AlexNet	0.953	0.948	0.845	0.856	0.795
3D map-AlexNet	0.964	0.951	8.821	0871	0.815
AU ONH-AlexNet	0.970	0.953	0.872	0.885	0839
ONH-VGG 16	0.962	0.954	0.858	0.881	0.801
3D map-VGG 16	0.981	0.966	0.894	0.924	0.854
AU ONH-VGG 16	0.991	0.979	0.907	0.943	0.887

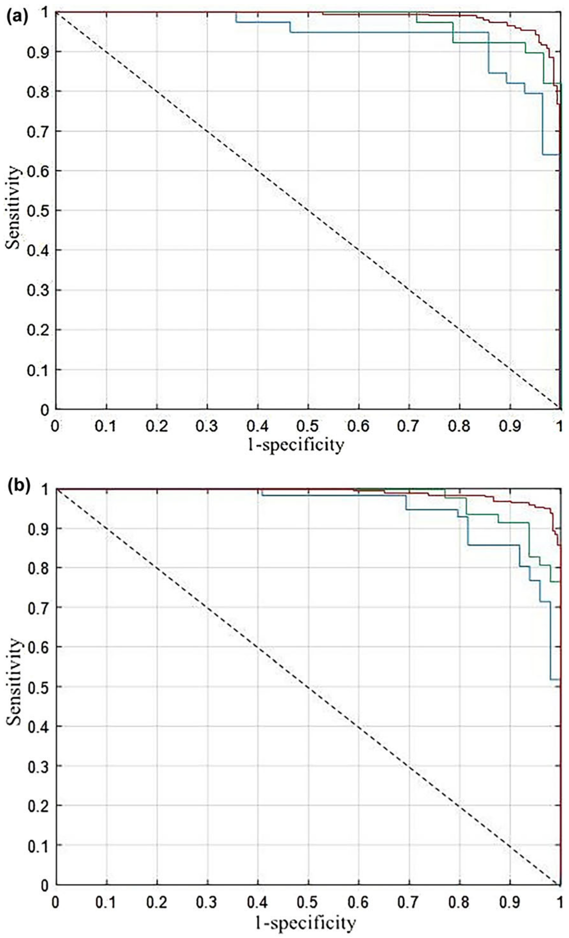

Fig.2 ROC curves of pretrained CNN models. (**a**) AlexNet; (**b**) VGG-16

transferred AlexNet. This may be benefited from more convolutional layers associated with the original VGG-16 network which help to extract more detailed features of fundus images. Meanwhile the image preprocessing approach including 3D augmentation also provide more data for training the parameters of fully connected layers and achieve better performance in differentiating the fundus images of glaucoma and non-glaucoma.

Figure 2 shows the ROC curves for pretrained AlexNet and VGG-16 models, respectively, and the blue/green/red curves reflect the results trained and tested from the ONH images / 3D map / augmented ONH images. According to the ROC curves and the calculated AUC results in Table 2, the neural network trained by the 3D topography dataset on the transferred AlexNet achieved a significant improvement compared to that trained on original ONH images and the average AUC approaches to 0.953 and 0.970, respectively. This result verifies that proper 3D reconstruction preprocessing can improve the classification performance of the network. The similar result was also demonstrated in the transferred VGG-16 and the average AUC approaches to 0.978 and 0.991, respectively. The results of ACC and Cohen's kappa value in Table 2 also confirmed this finding. The network trained by 3D topography map achieved the best classification performance with the better ACC and Cohen's kappa value which is almost equivalent to that using the expanded ONH images.

In comparison with AlexNet, the VGG-16 demonstrated a better performance in separating the fundus images of glaucoma from those of normal retinal conditions based on training augmented ONH images and has achieved ACC of 0.99 and Cohen's kappa of 0.88. There may be due to two main reasons. Firstly, the convolutional layers have potential advantage in extracting detailed features of the image. And more convolutional layer has, more useful features can be extracted from fundus images. Secondly, image augmentation not only increases the number of fundus images, but also corrects the network weights and bias in training process which improves the performance of differentiating the fundus images of glaucoma and non-glaucoma.

Conclusions

This paper exploited the usefulness of the augmented 3D fundus data for the Glaucoma-Net system transferred from pre-trained AlexNet and VGG-16 architectures. Experiments have been designed to verify the effectiveness of the different training data on the classification performance of the Glaucoma-Net system. The experimental results revealed that either expanded colour ONH fundus images or 3D topography maps of ONH images help to improve the classification performance, and the proposed Glaucoma-Net system is able to achieve better results compared to the state-of-the-art CNN classification systems. Consequently, the Glaucoma-Net system with suitable data preprocessing strategies can be adopted for the task of separating glaucoma and non-glaucoma fundus images so as to assist ophthalmologists in the diagnosis and treatment of glaucoma.

Author contributions P.P. Wang carried out the experiment and drafted the manuscript with support from all authors. J. Sun conceived the original project and M.Y. Yuan helped to supervise the project.

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Data availability The authors confirm that the data supporting the findings of this study are available within the articles [22–25] and their supplementary materials.

Compliance with ethical standards

Conflict of interest The authors declare that they have no competing interests.

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